

S&P 500 Market Direction Classification

A leakage-aware logistic regression case study

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1 Executive Summary

This case study tests whether a small set of lagged technical indicators can classify the next observed daily direction of the S&P 500. The workflow is designed to prevent look-ahead bias: each predictor uses information available no later than the prior trading day, and the final model is evaluated on the latest 20% of observations in chronological order.

The logistic regression achieved **49.6% test accuracy**, compared with a **51.4% majority-class baseline**. The model therefore did not demonstrate reliable out-of-sample improvement over a naive benchmark. This negative result is informative: simple lagged return, volatility, and moving-average signals are not sufficient by themselves for dependable daily direction forecasting over this test period.

2 Business Question

Can recent price behavior provide a useful, interpretable signal for whether the S&P 500 closes up or down on the next observed trading day?

The objective is classification, not causal inference or investment advice. A useful model should outperform a transparent baseline on future, unseen observations rather than only fitting historical data well.

3 Data and Feature Engineering

The dataset contains 2,518 daily observations from January 2014 through December 2023. The binary target equals one when the daily return is positive and zero otherwise.

Three predictors are constructed:

- **Lagged return:** the previous trading day's return.
- **Five-day volatility:** the rolling standard deviation of returns through the previous trading day.
- **Ten-day moving-average gap:** the prior close relative to its trailing ten-day moving average.

Lagging the rolling features is essential. Using the current day’s close or return to predict the current day’s direction would leak target information into the model and inflate performance.



Figure 1: S&P 500 closing price over the study period. The dashed line marks the chronological train-test split.

4 Modeling Strategy

The first 80% of observations (2,006 rows) form the training set, ending on January 03, 2022. The remaining 502 observations form the test set, running from January 04, 2022 through December 29, 2023.

The full logistic regression is

$$\log \left(\frac{P(Y_t = 1)}{1 - P(Y_t = 1)} \right) = \beta_0 + \beta_1 R_{t-1} + \beta_2 \sigma_{t-1,5} + \beta_3 G_{t-1,10}.$$

Candidate specifications are compared with AIC on the training data only. The full model is retained as the common reference specification for the held-out evaluation.

Table 1: Candidate-model comparison on the training sample.

Model	AIC	Delta AIC
Return + trend gap	2754.23	0.00
Return + volatility	2754.44	0.20
Full model	2756.21	1.97
Volatility + trend gap	2761.49	7.26

Table 2: Training-sample coefficients for the full logistic regression.

Term	Estimate	Std. error	p-value	95% CI lower	95% CI upper
Intercept	0.2042	0.0729	0.0051	0.0610	0.3469

Term	Estimate	Std. error	p-value	95% CI lower	95% CI upper
Lagged return	-0.1377	0.0519	0.0080	-0.2413	-0.0373
Five-day volatility	0.0108	0.0678	0.8730	-0.1210	0.1457
Ten-day moving-average gap	-1.7246	3.6057	0.6324	-8.7974	5.3600

5 Out-of-Sample Evaluation

Table 3: Performance on the chronological holdout set.

Metric	Value
Test accuracy	0.496
Majority-class baseline	0.514
Balanced accuracy	0.506
Brier score	0.258
Sensitivity	0.881
Specificity	0.132

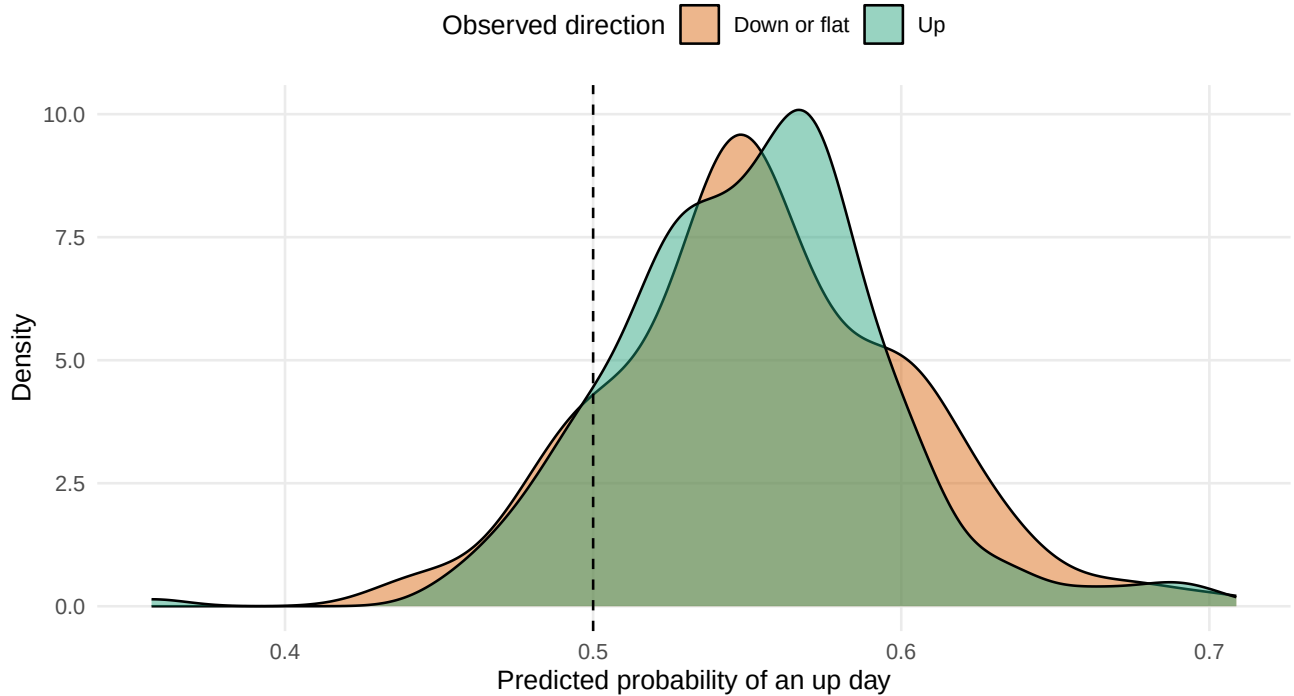


Figure 2: Predicted probability distributions by observed market direction in the test period.

The probability distributions overlap heavily, which is consistent with the model’s weak discrimination. Its **49.6% accuracy is below the 51.4% baseline**, and its Brier score is 0.258. The model should not be treated as a deployable trading signal.

6 Conclusions and Next Steps

The analysis provides a transparent example of building and testing a financial classification model without look-ahead bias. The lagged return has some training-sample association with market direction, but the combined feature set does not generalize well enough to beat a simple majority-class rule in the holdout period.

Useful next steps would include walk-forward validation, transaction-cost-aware decision rules, richer macroeconomic and market-microstructure features, regularization, and explicit probability calibration. Any future model should continue to be evaluated chronologically and against simple baselines.

This case study is for educational and portfolio purposes only and is not financial advice.